

# Harmonics and Algebraic Structure

Why the Same Mathematics — and What It Yields

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## Status markers

- [K]** *Confirmed* — numerically verified.
- [B]** *Proved* — algebraically established.
- [S]** *Speculative* — open.
- [E]** *Established* — textbook mathematics, not FFGFT-specific.
- [Q]** *Source* — external primary value.

## Abstract

The corpus describes harmonic structures with the algebra of finite fields in several places: commas and fractal correction (Doc. 060), overtones as torus spectrum (Doc. 159), the entry rule of harmonic primes and chords as partitions (Doc. 342), zeta hierarchy and resolution floor (Doc. 343), consonance as coupling geometry (Doc. 328). This document does not repeat those results. It answers two questions left open there: *why* the same mathematics describes both subjects — and *what that yields*.

### Results:

1. Theorem A **[B]**: In a finite field every “circle of fifths” closes exactly; a comma is impossible. The musical comma and the fractal correction  $K_{\text{frak}}$  are therefore not properties of the Galois structure but of the embedding into the continuum. This explains why  $K_{\text{frak}}$  becomes visible in the corpus only at the SI transition (Doc. 333).
2. Theorem B **[B]**: The two exceptions of the 5-limit reading in Doc. 060 (strange with prime 13, top with prime 7) are not exceptions in the Galois reading: 13 and 7 are the harmonic primes of depth  $k = 3$  and  $k = 6$  (Doc. 343 Theorem E). All nine Yukawa coefficients lie in the Galois grid  $\{2, 3, 5, 7, 11, 13\}$ ; the prime 11 ( $k = 5$ ) does not occur.
3. Observation C **[S]**: Three twelves coincide — twelve-tone temperament as the quotient of the Tonnetz by two commas,  $|(\mathbb{Z}/13)^*| = 12$  as the carrier group of the four fermion sectors,

and the Farey denominator height  $Q_c \approx 11\text{--}12.5$  at comma resolution. The mechanisms differ; the agreement in order is an observation, not a theorem.

4. Theorem D **[B]**: The answer to the why-question:  $\tilde{T} \cdot m = 1$  turns mass ratios into period ratios; period ratios are characters of cyclic groups; the  $Z_3$  reduction turns these into arithmetic in  $\text{GF}(3^k)$ . Harmonics and Galois classification are not analogous but two representations of the same object.

Verification script: `pruef_358_harmonik_algebra.py` (22/22).

## 1 Starting point: what the corpus already says

The following table assigns the existing results and is only cited below.

| Harmonics                                       | Algebra / FFGFT                           | Doc.     |
|-------------------------------------------------|-------------------------------------------|----------|
| Overtone series $f, 2f, 3f, \dots$              | Laplace spectrum on $S^1, T^d$            | 159      |
| Pythagorean comma +1.364 %                      | $K_{\text{frak}} = 1 - 100\xi, -1.333 \%$ | 060, 133 |
| 5-limit ratios                                  | Yukawa coefficients $r_i$                 | 006, 060 |
| Circle of fifths                                | torus topology, winding numbers           | 060, 189 |
| Temperament                                     | renormalisation / $K_{\text{frak}}$       | 060      |
| Consonance $\leftrightarrow$ denominator height | Farey height $Q_c$ , Arnold tongues       | 328, 343 |
| Harmonic primes 5, 7, 11, 13                    | entry depth $k = \text{ord}_p(3)$         | 342, 343 |
| Chords                                          | partitions of 8 in $G(6)$                 | 342      |
| 7-limit as hearing threshold                    | Euler-factor threshold $p^* \approx 9.76$ | 343      |
| Resolution limit $\approx 1 \%$                 | $100\xi$ as fractal step                  | 343      |

What remains open in all of these documents is why this assignment holds at all — whether it is analogy or identity.

## 2 Theorem A: There is no comma in a finite field

**Theorem A. [B]** Let  $q = 3^k$ . Every element of  $\text{GF}(q)^*$  has finite order dividing  $q - 1$ . Every chain  $g, g^2, g^3, \dots$  generated by an element  $g$  returns exactly to 1. A comma — a remainder after an almost-closed cycle — is impossible in  $\text{GF}(q)^*$ . In  $\mathbb{Q}^*$ , by contrast, no chain of fifths can close:  $3^a = 2^b$  has no solution  $a, b > 0$ .

*Proof.* The first part is Lagrange's theorem for the cyclic group  $\text{GF}(q)^*$  **[E]**. The second is unique factorisation in  $\mathbb{Z}$  **[E]**.

The content lies not in the proof but in the consequence for the corpus. Doc. 060 places the Pythagorean comma  $(3/2)^{12}/2^7$  and  $K_{\text{frak}} = 1 - 100\xi$  side by side and finds the near-exact cancellation 1.00013. Theorem A says *where* both quantities live: in  $\mathbb{Q}^*$  and  $\mathbb{R}$ , not in  $\text{GF}(27)^*$ . The Galois structure itself knows no remainder; all cycles close. The remainder appears only when the finite structure is mapped onto the number line — at the transition from winding numbers to measured masses in SI units.

This is exactly the position of Doc. 333:  $K_{\text{frak}}$  takes no part in the lossless pullback, sits in  $R_m$ , and becomes visible only at the SI transition. Theorem A supplies the algebraic reason. And it is the position of Doc. 343 Theorem G': the 1.05 % are a fractal step of the embedding, not a property of the Galois grid.

**Corollary A' [K]**: The cancellation comma  $\cdot K_{\text{frak}} \approx 1$  of Doc. 060 is thus not a coincidence of two independent systems but two measurements of the same embedding error — once from the ratio  $3/2$ , once from the ratio  $4/3$  (electron coefficient, Doc. 006). The residual 0.22 cent is the difference of the two starting ratios, not a third correction.

### 3 Theorem B: The exceptions of the 5-limit reading are Galois primes

Doc. 060 Table 2 classifies the nine Yukawa coefficients of Doc. 006: seven are 5-limit  $(2, 3, 5)$ , two are not — strange  $26/9$  carries the prime 13, top  $1/28$  carries the prime 7. Doc. 060 marks both as exceptions without musical interpretation.

Doc. 343 Theorem E shows independently: the prime  $p$  enters the  $\text{GF}(3^k)$  hierarchy exactly at  $k = \text{ord}_p(3)$ , and for  $k \leq 6$  these are precisely 13 ( $k = 3$ ), 5 ( $k = 4$ ), 11 ( $k = 5$ ), 7 ( $k = 6$ ).

**Theorem B. [B]** All nine Yukawa coefficients of Doc. 006 have prime factors exclusively in  $\{2, 3, 5, 7, 11, 13\}$ . The two non-5-limit coefficients carry exactly the primes of Galois depth  $k = 3$  (strange: 13) and  $k = 6$  (top: 7). The prime 11 ( $k = 5$ ) occurs in no coefficient.

*Proof.* Direct factorisation of the nine fractions (script B1–B5).

Three remarks:

1. The 5-limit reading of music is the Galois reading up to depth  $k = 4$  (primes 2, 3, 5; the prime 13 does enter at  $k = 3$ , but as the group order  $|Z_{13}|$ , not as a coefficient). What appears in music as “beyond the 5-limit” is algebraically the next layer. The exceptions of Doc. 060 are thereby resolved.
2. Strange carries the prime 13 — the same depth  $k = 3$  that carries the lepton orbits  $(\text{GF}(27)^* \cong Z_2 \times Z_{13})$ , Doc. 339). Top carries 7 and needs  $k = 6$ , i.e.  $\text{GF}(729)$  — exactly the candidate that R115 names for the CKM/PMNS angles.
3. Whether the quark coefficients are geometric or calibrated remains open after Doc. 289 **[S]**. Theorem B says only: *if* they are geometric, they lie in the right grid. That is compatibility, not proof.

**Falsifiable consequence [K]:** A fundamental mass coefficient with a prime factor  $> 13$  (or with 11) would be incompatible with Doc. 343 Theorem E. The prediction is testable on every coefficient derived in future.

### 4 Observation C: Three twelves

Three quantities of order 12 occur, from three different mechanisms:

1. **Temperament.** The 5-limit Tonnetz is the free lattice  $\mathbb{Z}^2$  in the exponents of 3 and 5 (modulo octave). Twelve-tone temperament is the quotient by the syntonic comma  $81/80 \mathbb{H}(4, -1)$  and the diesis  $128/125 \mathbb{H}(0, -3)$ :

$$|\mathbb{Z}^2 / \langle (4, -1), (0, -3) \rangle| = |\det| = 12.$$

The 12 arises by *quotient* of a torsion-free lattice **[E]**.

2. **Galois.**  $|\langle \mathbb{Z}/13 \rangle^*| = 12$ ; the element 3 has order 3 in it (Doc. 343 Theorem E), the four cosets of  $\langle 3 \rangle$  are the four fermion sectors, their size 3 the three generations (Doc. 343 Theorem D, Doc. 348). Here the 12 is *intrinsic torsion*: the group is cyclic, no quotient needed.
3. **Resolution.** The Farey denominator height  $Q_c = \pi/\sqrt{6\varepsilon}$  (Doc. 343 Plain Language II) gives, at comma resolution  $\varepsilon \in \{1.36\%, 1.33\%, 1.05\%\}$ , the values  $Q_c \approx 11.0, 11.1, 12.5$ . Denominators above  $\approx 12$  are indistinguishable at this resolution.

**Observation C. [S]** The three twelves come from three different mechanisms (quotient, torsion, resolution limit). That they agree in order is an observation. A theorem mapping the three mechanisms onto one another is not available.

A hint that supports the observation without proving it: twelve-tone temperament is not convention but the smallest quotient at which  $3^{12} \approx 2^{19}$  closes to comma accuracy — i.e. the smallest order at which the free lattice *imitates* the torsion of the finite field. In this sense temperament is an approximation of the Galois structure by the continuum, and Theorem A says what is lost: exact closure.

## 5 Theorem D: Why the same mathematics

The question whether the assignment in the table of §1 is analogy or identity has a precise answer in three steps, all established in the corpus:

1.  $\tilde{T} \cdot m = 1$  (Doc. 003, 010): mass is inverse period. A mass ratio  $m_i/m_j$  is a period ratio  $\tilde{T}_j/\tilde{T}_i$  — a frequency ratio. This is not a metaphor but the definition of the quantity.
2. Frequency ratios on a periodic structure are characters of the cyclic group  $S^1$ ; on  $T^4$  they are the winding lattice  $\mathbb{Z}^4$  (Doc. 159, 189). The overtone series is the character spectrum of  $S^1$ ; the mass spectrum is the character spectrum of  $T^4$ .
3. The reduction mod 3 (Doc. 336 ring homomorphism  $\varphi(N) = N \bmod 3$ , Doc. 341) maps the winding lattice to  $\text{GF}(3^k)$ . The  $\mathbb{Z}_3$  orbifold action is a second, different  $\mathbb{Z}_3$ : the reduction  $N \bmod 3$  is an additive triality, invariant under the cyclic permutation; the orbifold action fixes the sectors  $\mathcal{H}_k$  through its eigenphase  $\omega^k$  (Doc. 321, 336). The Frobenius action  $x \mapsto x^3$  is the sector pairing  $k \leftrightarrow -k$  (Doc. 336 main result).

**Theorem D. [B]** Harmonic analysis of a mass spectrum and Galois classification in  $\text{GF}(3^k)$  are two representations of the same object — the character spectrum of  $T^4/\mathbb{Z}_3$ . The first reads it in  $\mathbb{Q}^* \subset \mathbb{R}$  (frequency ratios), the second in  $\text{GF}(3^k)^*$  (orbits). The passage between them is reduction mod 3 in one direction and embedding into the continuum in the other; Theorem A describes what the embedding adds (the comma), Doc. 333 describes what the reduction removes (nothing — the pullback is lossless).

*Proof.* Concatenation of the three cited steps; no new proof step.

The assignment in §1 is therefore not an analogy. It is the translation table between two representations.

## 6 What the connections yield

The question “what does this give us” has five answers, each with a concrete corpus return:

1. **An admissibility criterion instead of curve fitting.** Which primes may occur at all in a fundamental ratio is fixed algebraically (Doc. 343 Theorem E). Theorem B applies this to Doc. 006 and finds all nine coefficients in the grid. A fit would not have guaranteed this; the algebra forces it.
2. **A built-in resolution limit.** Harmonics has known for centuries that consonance ends at about the 7-limit. Doc. 343 Theorem F gives the number ( $p^* \approx 9.76$ ), Theorem G’ the reason (100%). For mass spectroscopy this means: ratios with denominators  $\geq 12$  are indistinguishable from random products (Doc. 342 Theorem D). Harmonics supplies the intuition, algebra the limit, and both prevent over-interpretation — the look-elsewhere error.

- 3. A reason for the existence of commas.** Theorem A: the comma is the price of embedding.  $K_{\text{frak}}$  is not an extra parameter but the necessary difference between exact closure in  $\text{GF}(27)^*$  and the number line. Doc. 060 found the agreement; Theorem A says why it is no coincidence.
- 4. A physical reading of consonance.** Doc. 328 shows mixing angles as coupling degrees and Arnold tongues as stability regions. Consonance in this reading is not aesthetics but the width of an Arnold tongue — and conversely the mixing angles of the Standard Model are “intervals” with determinable consonance. This is the direction in which the CKM/PMNS values (R115, **[S]**) are to be approached: not as numbers but as denominator heights.
- 5. A working intuition that holds.** The Euler Tonnetz was the practical precursor of  $\xi$  (Doc. 060, 009). Theorem D says why the precursor held: it was not an analogy but already the continuum representation of the object that the Galois representation later made exact. Whoever thinks in the Tonnetz thinks in  $T^4/Z_3$  — with comma.

## 7 Scope

- This document derives no masses and changes no numerical value in the corpus.
- Observation C is marked **[S]** and remains so until a theorem connects the three mechanisms.
- The quark coefficients remain calibration values after Doc. 289; Theorem B establishes compatibility, not origin.
- The connection to auditory physiology (why the ear stops at the 7-limit) is not a subject here; Doc. 343 Theorem F treats the mathematical limit, not the biological one.

## 8 Summary

| Thm. | Statement                                                               | Status     |
|------|-------------------------------------------------------------------------|------------|
| A    | No comma in a finite field; comma = embedding error                     | <b>[B]</b> |
| A'   | comma $\cdot K_{\text{frak}} \approx 1$ = two measurements of one error | <b>[K]</b> |
| B    | All nine Yukawa coefficients in the Galois grid; 13, 7 no exceptions    | <b>[B]</b> |
| C    | Three twelves (quotient / torsion / resolution)                         | <b>[S]</b> |
| D    | Harmonics and Galois: two representations of one object                 | <b>[B]</b> |

## Verification scripts

2/python/Dok358\_Skripte/pruef\_358\_harmonik\_algebra.py — 22 assertions: A1 ( $\text{GF}(3^k)^*$  explicit,  $k = 1..6$ ), A2 ( $\mathbb{Q}^*$ ), A3 (comma/ $K_{\text{frak}}$ ), B1–B5 (prime factors of the nine coefficients), C1–C4 (three twelves), D1–D2 (limit thresholds).

## **Note on the use of AI assistance**

AI assistance (Claude, Anthropic) was used for wording, LaTeX typesetting and the creation of the verification script. The substantive claims, the status assignment and the responsibility for the text lie with the author.